

Irreversible Metadata Scrubber — Progress & Testing Report

Phase 1: Images (JPEG and PNG) · Author: Mohammad Ghorayeb · Date: 2026-07-23

1. What this project is (in one paragraph)

Photos and files carry hidden information called **metadata** — where a photo was taken (GPS), which phone took it, the date, and sometimes a small copy of the image (a **thumbnail**). This project builds a tool that **removes that hidden information so it can never be recovered again** — not even by someone opening the file byte-by-byte with forensic tools. The key word is **irreversible**: not just hidden or deleted, but truly gone from the file.

The goal is a **working tool**, tested against a realistic attacker. That attacker is a **journalist or amateur investigator with off-the-shelf tools** — not a nation-state. We call this the "medium-tier" adversary.

2. The two kinds of snoop we defend against

We split the attacker into two levels, because beating one is easy and beating the other is hard.

Name	Who they are	What they try to read
A1	The metadata snoop	Opens the file and reads the tags: GPS, camera, date, thumbnail, comments.
A2	The fingerprint snoop	Even with all tags gone, looks at <i>*how the file was compressed*</i> to guess which program or phone made it.

A1 is about content you can read. A2 is about a hidden "signature" left by the compressor — like recognising a typewriter from the shape of its letters, even with no name on the page.

3. What we built

The tool auto-detects the file type (by its real contents, not its name) and strips it. We built two formats so far, each with several **cleaning levels** called **fidelity tiers**:

Level	Plain meaning	Cost
F1	Delete all metadata, don't touch the picture at all (byte-for-byte identical pixels).	None
F2	Delete all metadata and repack the compression cleanly. Picture still pixel-perfect.	None
F3	Fully re-compress the picture with one standard compressor, so every file looks the same.	Tiny, invisible quality loss

- **JPEG** (photos): F1, F2, and F3 all built.
- **PNG** (screenshots, graphics): F1 and F2 built. PNG needs no F3 — it is a lossless format, so F2 already does the strongest job with **zero** quality loss.

Every tier is **fail-closed**: if the tool cannot guarantee a clean result, it refuses and writes nothing, rather than quietly producing a file that **looks** clean but isn't.

What gets removed

EXIF, GPS, camera make/model, timestamps, software version, XMP, Extended XMP, IPTC, ICC colour profiles, embedded thumbnails, comment blocks, and any extra bytes tacked onto the end of the file (the "trailer"). The tool also makes sure it leaves **no signature of itself** — no "made by our scrubber" mark.

4. How we test — "differential testing"

The core method is simple and powerful. Take **two files with the same picture but different metadata**. Scrub both. If the results are not identical, the leftover difference **is** a leak — it can only be the metadata that failed to disappear.

We built an automated **test harness** that does this at scale, plus a battery of unit tests. As of this report: **127 automated tests, all passing**.

5. The fingerprint experiment (the interesting part)

To test the A2 fingerprint snoop, we ran a real experiment based on published forensic research (Kornblum).

Setup. We took real photos and compressed each one with **three different compressors** (Apple's, libjpeg-turbo, and ffmpeg). Same picture, different program. Then we checked the hidden compression signature — for JPEG this lives in the **quantization tables (DQT)**.

Finding. The signature was **perfectly consistent for each compressor and different across them** — a reliable fingerprint. And crucially, it **survived F1 and F2 scrubbing unchanged**. Removing metadata does **not** remove the fingerprint.

The fix. F3 re-compresses everything with one standard compressor, so all files end up with the **same** signature — the fingerprint becomes useless. For PNG, F2 does the same thing losslessly.

6. Results — the scoreboard

Every cell below was **measured on real files**, not assumed.

Format	A1 metadata	A1 metadata	A1 metadata	A2 fingerprint	A2 fingerprint	A2 fingerprint
	F1	F2	F3	F1	F2	F3
JPEG	PASS	PASS	PASS	fail	fail	PASS
PNG	PASS	PASS	n/a	fail	PASS	n/a

Reading it in plain words:

- Against the **metadata snoop (A1)**, the tool **wins everywhere**.
- Against the **fingerprint snoop (A2)**, you need the strongest mode: **JPEG F3** (tiny quality cost) or **PNG F2** (no cost at all).
- **PNG F2 is the standout result:** untraceable **and** pixel-perfect — something JPEG cannot do, because a JPEG's fingerprint is baked into the pixels while a PNG's is not.

7. The big test — can it be reversed?

We built a dedicated **adversarial recovery test** that plays the attacker. We planted **known secret words** in every metadata hiding spot, scrubbed the file, then attacked the scrubbed bytes with real forensic techniques:

1. Raw byte search for the secret words (normal and wide-character text).
2. Hunting for embedded thumbnails (a hidden second image).
3. Checking for extra bytes at the end of the file.
4. Running **ExifTool** (the industry-standard metadata reader).

We ran this on a synthetic "torture" file stuffed with every kind of metadata, **and on a real iPhone photo**.

Results on a real iPhone photo

Check	Before scrub	After scrub (any level)
Metadata tags (ExifTool)	151	0
GPS location	33°N, 35°E	gone
Camera / model / date	iPhone 16, full date	gone
Embedded thumbnail	present	gone
Leftover secret text	found	none

Nothing we scrubbed could be recovered. This test is now permanent — it runs automatically and will catch any future change that reintroduces a leak.

8. Could someone working at the binary level restore it?

This is the honest, important answer.

For the metadata itself — no. We do not **hide** or **blank** the data; we **physically remove those bytes and never write them into the new file**. An investigator opening the scrubbed file in a hex editor has nothing to carve back, because the GPS, camera name, and thumbnail are simply **not in the file anymore**. You cannot recover bytes that do not exist. This is the real meaning of **irreversible**.

What a clever analyst can still learn (and why it is not a metadata leak):

- **The compressor fingerprint (at F1/F2 only).** The file may still hint "an Apple encoder made this." That is **not** your GPS or your name — it only narrows the **type of device**. **F3 (JPEG) and F2 (PNG) erase even this.**
- **Sensor noise (PRNU).** Every camera sensor leaves a faint, unique noise pattern **in the pixels themselves**. With the original camera in hand, an expert could statistically link a photo to it. This is a **known limit of every tool** — it lives in the image, not the metadata, and removing it would destroy the picture.
- **"Was re-saved" traces (after F3).** Faint statistical hints that the image was compressed before. This reveals **that** it was processed, never **what** the metadata said.

We **document** these residuals openly rather than pretend they do not exist — which is the scientifically honest thing to do.

9. What is done and what is left

Done:

- JPEG scrubber — F1, F2, F3.
- PNG scrubber — F1, F2.
- Automated differential-testing harness + 127 passing tests.
- The DQT fingerprint experiment (A2), measured on real photos.
- A permanent forensic-recovery / irreversibility guard.

Left (future phases):

- More formats: audio (MP3), documents (PDF, Word), video (MP4), camera RAW.
- A side-by-side comparison against existing tools (MAT2, ExifTool) — evidence for the funding proposal.
- Two small optional image loose ends (a formal PNG results table, and hardening one edge of the fingerprint check).

10. Summary

We have a **working, tested tool** for the two most common image formats. It **defeats the metadata snoop completely**, and **defeats the fingerprint snoop** at its strongest settings — losslessly for PNG. Most importantly, the information it removes is **forensically unrecoverable**, because it is deleted from the file itself, not merely hidden — proven by attacking a real iPhone photo with the same tools an investigator would use. The remaining traces (device fingerprint, sensor noise) are **documented honestly** and are the known limits of the entire field, not shortcomings of this tool.